

# CNC Services Northwest

*Centroid CNC control sales, service, training and support*

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These are some of the rotary position encoders used on Centroid controls over the years.

## Dynapar F14 and F18 Encoders

Made by Dynapar division of Danaher Controls.

[Manufacturer's web site](#)



1/4", 3/8" and 6mm thru-hole hollow-shaft, flex mount, 2x set screws in collar.

Used on various motors and as replacement, ca. 2003-present.

Set screws are 90° apart, and accept 1/16" hex wrench. Flexible base is anchored with two screws.

Motors supplied by the Centroid factory with these encoders use a black plastic Molex SL Modular series connector.

When used as a replacement for the EPC encoders in Servo Dynamics/SEM motors on Revolution machines, a DE9 male connector is fitted.

These encoders are generally available for immediate shipment.

*Dynapar F14 or F18 encoder with 6mm, 1/4" or 3/8" bore: \$240*

## Pin-outs

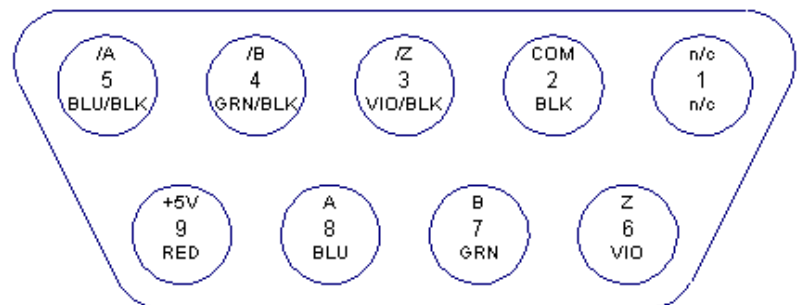
### Molex SL Modular (Centroid)

| Pin | Signal | Color       |
|-----|--------|-------------|
| 1   | ch A   | BLU         |
| 2   | ch /A  | BLU/<br>BLK |
| 3   | ch B   | GRN         |
| 4   | ch /B  | GRN/<br>BLK |
| 5   | ch Z   | VIO         |
| 6   | ch /Z  | VIO/<br>BLK |
| 7   | Vcc    | RED         |
| 8   | COM    | BLK         |
| 9   | shield | shield      |

### DE9M (Revolution)

| Pin   | Signal | Color       |
|-------|--------|-------------|
| 1     | n/c    |             |
| 2     | COM    | BLK         |
| 3     | ch /Z  | VIO/<br>BLK |
| 4     | ch /B  | GRN/<br>BLK |
| 5     | ch /A  | BLU/<br>BLK |
| 6     | ch Z   | VIO         |
| 7     | ch B   | GRN         |
| 8     | ch A   | BLU         |
| 9     | Vcc    | RED         |
| shell | shield | shield      |

Dynapar F14, DE9 for Revolution  
(back of male plug)



## Quantum Devices QD145

Distributed by Quantum Devices, Barneveld WI

[Manufacturer's web site](#)



1/4" thru-hole hollow-shaft, flex mount, 2x 4-40 set screws in collar (use 0.050" hex wrench).  
Used on black Glentek DC servo motors ca. 2005.

Also available as a retrofit encoder with 3/8", 6mm, 8mm, 10mm and other bore diameters.

The electronic connection is a black plastic Molex SL Modular series connector.

## Pin-out

| Pin | Signal | Color |
|-----|--------|-------|
| 1   | ch A   | BRN   |
| 2   | ch /A  | WHT   |
| 3   | ch B   | BLU   |

| Pin | Signal | Color  |
|-----|--------|--------|
| 4   | ch /B  | GRN    |
| 5   | ch Z   | ORG    |
| 6   | ch /Z  | YEL    |
| 7   | Vcc    | RED    |
| 8   | COM    | BLK    |
| 9   | shield | shield |

## Quantum Devices QR12

Distributed by Quantum Devices, Barneveld WI

[Manufacturer's web site](#)







1/4", 3/8", etc. thru-hole hollow-shaft, flex mount, 2x 4-40 set screws in collar.  
Used as field replacement, ca. 2012-present.

The electronic connection is a black plastic Molex SL Modular series connector.

These encoders are generally available for immediate shipment.

*Quantum QR12 encoder with 1/4" or 3/8" bore: \$240*

## Pin-outs

### Molex SL Modular (Centroid)

| Pin | Signal | Color  |
|-----|--------|--------|
| 1   | ch A   | BRN    |
| 2   | ch /A  | WHT    |
| 3   | ch B   | BLU    |
| 4   | ch /B  | GRN    |
| 5   | ch Z   | ORG    |
| 6   | ch /Z  | YEL    |
| 7   | Vcc    | RED    |
| 8   | COM    | BLK    |
| 9   | shield | shield |

### DE9M (Revolution)

| Pin   | Signal | Color  |
|-------|--------|--------|
| 1     | n/c    |        |
| 2     | COM    | BLK    |
| 3     | ch /Z  | YEL    |
| 4     | ch /B  | GRN    |
| 5     | ch /A  | WHT    |
| 6     | ch Z   | ORG    |
| 7     | ch B   | BLU    |
| 8     | ch A   | BRN    |
| 9     | Vcc    | RED    |
| shell | shield | shield |

Note that Centroid's factory pigtail for SL Modular (C-Grid) connectors has the A channel (BRN & WHT) and B channel (BLU & GRN) each reversed. The net result of this is exactly the same as if they were wired per the table above, which is based on Quantum's spec sheet.

## EPC 755A Encoder

Made by Encoder Products Company, Sandpoint Idaho.

[Manufacturer's web site](#)



1/4" and 3/8" blind hollow-shaft, flex mount, 2x set screws in collar.  
Used on orange-cap SEM DC servo motors ca. 1993-1999.

Set screws are 90° apart, and accept 1/16" hex wrench. Flexible base is anchored with two screws.

Orange-cap SEM motors supplied by the Centroid factory from 1994 to 1998 or so used these encoders with a 3/8" bore and a black plastic Molex SL Modular series connector.

Orange-cap SEM motors with a Servo Dynamics label, supplied with most Revolution machines from 1996 to 1999, used these encoders with a 1/4" bore and a DE9 male connector (as pictured).

These encoders are still available through EPC distributors. However, because of their high cost I usually replace failed units with an equivalent Dynapar or Quantum encoder instead.

## Pin-outs

### Molex SL Modular (Centroid)

| Pin | Signal | Color  |
|-----|--------|--------|
| 1   | ch A   | BRN    |
| 2   | ch /A  | YEL    |
| 3   | ch B   | RED    |
| 4   | ch /B  | GRN    |
| 5   | ch Z   | ORG    |
| 6   | ch /Z  | BLU    |
| 7   | Vcc    | WHT    |
| 8   | COM    | BLK    |
| 9   | shield | shield |

### DE9M (Revolution)

| Pin   | Signal | Color  |
|-------|--------|--------|
| 1     | n/c    |        |
| 2     | COM    | BLK    |
| 3     | ch /Z  | BLU    |
| 4     | ch /B  | GRN    |
| 5     | ch /A  | YEL    |
| 6     | ch Z   | ORG    |
| 7     | ch B   | RED    |
| 8     | ch A   | BRN    |
| 9     | Vcc    | WHT    |
| shell | shield | shield |

## Sumtak LHE-514 Encoder







6mm blind hollow-shaft, flex mount, 2x set screws in collar.  
Used on blue-cap SEM DC servo motors ca. 2001-2003.

Set screws are 90° apart, and accept 1.5mm hex wrench. Flexible base is anchored with a single screw.

This encoder is no longer available, but I have equivalent Dynapar encoders in stock.

### Pin-out

| Pin | Signal | Color  |
|-----|--------|--------|
| 1   | ch A   | RED    |
| 2   | ch /A  | PNK    |
| 3   | ch B   | GRN    |
| 4   | ch /B  | BLU    |
| 5   | ch Z   | YEL    |
| 6   | ch /Z  | ORG    |
| 7   | Vcc    | WHT    |
| 8   | COM    | BLK    |
| 9   | shield | shield |

### Danaher M21 Encoder





3/8" and 6mm thru-hole hollow-shaft, direct mount.  
Used on blue-cap SEM DC servo motors ca. 1999-2000

The installation procedure is tricky, and should not be attempted without manufacturer instructions. The encoder body rotates to capture or free the encoder disk. You install the encoder with the disk captured; tighten the shaft clamp with a 2mm (5/64") hex wrench; then rotate the body to free the disk. To remove the encoder, you rotate the body to capture the disk; loosen the shaft clamp with a 2mm or 5/64" hex wrench; then remove the base.

#### [Danaher M21 Installation Guide](#)

This encoder is still available, but I generally use an equivalent Dynapar F-series encoder as a replacement because of the simpler installation procedure.

## Pin-outs

### Molex SL Modular

| Pin | Signal | Color   |
|-----|--------|---------|
| 1   | ch A   | GRN     |
| 2   | ch /A  | RED/BLK |
| 3   | ch B   | ORG     |
| 4   | ch /B  | WHT/BLK |
| 5   | ch Z   | WHT     |
| 6   | ch /Z  | BLU     |
| 7   | Vcc    | RED     |
| 8   | COM    | BLK     |
| 9   | shield | shield  |

### Danaher 10-pin

| Pin | Signal | Color   |
|-----|--------|---------|
| 1   | n/c    |         |
| 2   | Vcc    | RED     |
| 3   | COM    | BLK     |
| 4   | n/c    |         |
| 5   | ch /A  | RED/BLK |
| 6   | ch A   | GRN     |
| 7   | ch /B  | WHT/BLK |
| 8   | ch B   | ORG     |
| 9   | ch /Z  | BLU     |
| 10  | ch Z   | WHT     |

## Lucas Ledex S-10285

Distributed by Lucas Ledex, Vandalia OH

[Ledex](#) is evidently no longer in the optical encoder business.





3/8" blind hollow-shaft, flex mount, pinch screw in collar.  
Used on orange-cap SEM DC servo motors ca. 1994-1995.

Many orange-cap SEM motors supplied by the Centroid factory in 1994 and 1995 used these encoders with a 3/8" bore, but on a 1/4" stub shaft. Centroid installed a plastic bushing to adapt the shaft to the bore. This arrangement was prone to slippage if not adequately tightened, leading to gradual position loss issues.



The electronic connection was a black plastic Molex SL Modular series connector.

These encoders are no longer available. They can be easily replaced with a 1/4" Dynapar F14 or Quantum QR12, eliminating the need for the bushing.

### Pin-out

| Pin | Signal | Color   |
|-----|--------|---------|
| 1   | ch A   | WHT     |
| 2   | ch /A  | BLU     |
| 3   | ch B   | GRN     |
| 4   | ch /B  | ORG     |
| 5   | ch Z   | WHT/BLK |
| 6   | ch /Z  | RED/BLK |



| Pin | Signal | Color  |
|-----|--------|--------|
| 7   | Vcc    | RED    |
| 8   | COM    | BLK    |
| 9   | shield | shield |

## Encoder Phasing and Direction

Most encoders follow a standard convention for phasing and rotation:

- A leads B for CW rotation as viewed from the shaft end
- B leads A for CCW rotation as viewed from the shaft end

For DC servo motors and for third-party drives, Centroid considers positive rotation (positive-going encoder counts) to be CCW rotation as viewed from the shaft end.

Centroid boards which receive the encoder input via 9-pin D-sub connectors (DE9, commonly misnamed as DB9) will count positive when the channel which is connected to pins 5 and 8 leads the channel which is connected to pins 4 and 7.

Therefore, when using any encoder which follows the industry convention, channel B should be wired to pins 5 and 8, and channel A should be wired to pins 4 and 7.

Centroid factory system schematics generally show channel B going to pins 5 and 8, and channel A going to pins 4 and 7. If you follow this guidance, you will generally get the correct results.

Centroid component hardware manuals (e.g. for the Oak, All-in-one DC, and MPU11 boards) have the A and B channel labels reversed. Evidently the electrical engineers, in their internal documentation, consider A leading B to be "positive", in spite of the conflicting convention out at the motor. Beware of this when referring the component manuals.

Revolution Machine wired their own DC servo motors and encoders, and they adopted a convention opposite that of the Centroid factory. Therefore when installing a standard encoder on a Revolution machine, you need to wire channel A to pins 5 and 8, and channel B to pins 4 and 7.

For their own AC-servo systems (AC/DC drives; SD3 and SD1 drives; and the SERVO4 drive) Centroid also adopted a reversed convention: positive motor rotation is CW as viewed from the shaft end of the motor.

[Why is the encoder counts/rev value 4x the encoder line count?](#)

## Other Encoder Trivia

Dynamics Research Corp. (DRC) made several families of encoders.

DRC 152-series encoders, sometimes found in SEM motors on Milltronics machines, are "differential open collector". These encoders have complementary channels (e.g. A, /A, B, /B, etc.) but they are NPN open-collector signals instead of line driver signals. As such, the encoder does not drive "high" lines high when they are not pulled low. While it would probably be possible to use these encoders by adding pullup resistors to each channel, it is better to just replace them.

The ones found on Milltronics machines are standard 1.5" solid-shaft encoders, some with 6mm shafts, and some with 1/4" shafts.

DRC F21-series encoders, by contrast, do have line driver output, and are therefore compatible with Centroid controls.

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*Last updated 23-Feb-2023 MBL*